NOW CYBER KINETIC WARFARE

The coordinated pager attacks that struck the Hezbollah raise the spectre of cyber kinetic warfare, which combines communication technologies with physical destruction. India faces a heightened risk as most of its connected devices depend on Chinese and foreign components

By Pradip R. Sagar

Illustration by **NILANJAN DAS/AI**

On September 17, around 3.30 pm Beirut time, thousands of pagers used by members of the Hezbollah, the Shia militant group, started exploding across Lebanon. The explosions killed 11 and injured 2,700. The next day, it was the turn of walkie talkies they used to blow up, killing over 20 and injuring 450 people. The mass electronic sabotage, presumed to have been orchestrated by the Israel Defense Forces (IDF) along with the country's spy agency, Mossad, represents a new chapter in electronic warfare. The pager/ walkie talkie attacks raise the spectre worldwide of the potential widespread weaponisation and targeting of personal gadgets. Though putting explosives in phones and detonating them is an old art in the secret agent's toolbox, what makes the alleged Israeli attack unique is the deep supply chain interdiction of a large cache of devices and the planting and detonating of explosives in so many of them at once. Meanwhile, the latest instalment of Israeli action against Hezbollah is being pursued through more conventional means and has taken on a dangerous dimension with the killing of Hezbollah's chief, Hasan Nasrallah, in an Israeli airstrike on September 27. In response to Nasrallah's death and the Israeli army's ground offensive against Hezbollah into Lebanon itself, Iran, Israel's main adversary in the region, launched around 200 ballistic missiles into Israel on October 1. Aimed at air bases and military installations, most were intercepted by Israel's Iron Dome air defence system.

The reason why military experts are terming the pager attacks as a watershed moment in electronic sabotage is the potential of such attacks on larger targets connected to the internet—say, civilian and military communications systems, or power grids—that can paralyse a country at war. The keyword here is not just cyber warfare, but cyber kinetic warfare. While cyber attacks involve an attempt to damage another nation's computers or information networks through viruses or denial-of-service attacks, cyber kinetic attacks use cyber attacks to inflict kinetic (or physical) damage on infrastructure or cause injury or death to people. The growth of cyber physical systems in everything, from automobiles, aircraft, personal gadgets and household appliances (the Internet of Things) to large national/ military assets, opens up frightening possibilities of acts of sabotage with both physical and cyber elements.

India is particularly vulnerable to such attacks, given its over-reliance on Chinese equipment in communication networks and Chinese-made surveillance cameras. Though concerns about Chinese surveillance and possible system failure triggered by embedded malware are being acted upon, the use of previously sabotaged hardware—like the pagers—being activated by a cyber attack or a malware that is programmed to trigger destruction, cannot be ruled out now.

"Terrorists can use weaponised electronics but only states have the capability to carry out the kind of large-scale attacks or mass surveillance that impacts national security," says Lt Gen. D.S. Hooda, former commander of the army's Northern Command. He says that while cyber attacks and surveillance are nothing new, the addition of a kinetic element is a novelty.

THREATS TO INDIA

Indian intelligence agencies have been issuing warnings about the country's communications network, whose hardware and software are largely of Chinese or foreign origin and thus can potentially be used to snoop on India during normal times and for cyber attacks in a future conflict. Indian telecommunications companies, like the state-owned Bharat Sanchar Nigam Limited (BSNL) and its subsidiary, Mahanagar Telephone Nigam Limited (MTNL) as well as Bharti Airtel and Vodafone Idea, are dependent on Chinese equipment. In 2020, the Centre mandated that telcos should use equipment only from vendors that have received "trusted source" certification from the government. The Chinese vendors Huawei and ZTE have yet to receive this certification. "Unless we have our own communications network using indigenous material, no one is safe. BSNL networks are based on Chinese equipment," says Colonel